

CHING CHAO

AI Research Engineering • Multi-Agent Systems • Quantitative Finance

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RESEARCH & DOMAIN PROFILE

Independent AI researcher and research engineer with advanced ML training and a career in institutional quantitative finance, portfolio management, and risk research. Builds multi-agent systems and controlled evaluation harnesses for high-stakes decisions under uncertainty. Research interests include autonomous scientific agents, model and evidence verification, representation adaptation, and robust decision-making under distribution shift.

SELECTED AI RESEARCH & ENGINEERING PROJECTS

Institutional Risk Agent Fleet | github.com/mbzuaiciao/institutional-risk-agent-fleet

- Built a hackathon multi-agent risk-investigation prototype with Gemini / Google ADK: specialist investigators, an adversarial challenger, deterministic claim verification, and human approval.
- Implemented Cloud Run, Pub/Sub, Firestore, role-bound permissions, audit trails, and a demo; 66 tests reported. Synthetic data; not a deployed trading or investment system. Devpost: devpost.com/software/institutional-risk-agent-fleet

Institutional Investment Agents | github.com/mbzuaiciao/institutional-investment-agents

- Studied whether specialist agents, critique, and verification improve research quality rather than merely response complexity; evaluated architectures in controlled synthetic tasks.
- Completed a 324-run model-capability × harness-capability simulator factorial; developed a real-model validation protocol (empirical real-model results pending).

AFML Research Harness | github.com/mbzuaiciao/afml-research-harness

- Built a zero-trust research evaluator with leakage controls, multiple-testing corrections, reproducible reporting, and provenance; EXP-011 probes adaptive-search overfitting.
- Completed deterministic R0/R1 baseline; self-hosted external-model confirmatory testing remains in progress.

SELECTED APPLIED AI WORK

G42 | Agentic AI for Institutional Investment Research

- Designed and prototyped an agentic fixed-income research workflow for a UAE sovereign wealth fund, combining retrieval, structured reasoning, memory, routing, tools, and traceable evidence.

K2 Risk | AI-Assisted Multi-Asset Risk Interpretation

- Explored LLM-assisted interpretation of quantitative risk models, scenarios, uncertainty, and model assumptions, preserving provenance and human decision authority.

SELECTED FINANCE CAREER HISTORY

LetterOne Group — Portfolio Manager: Global macro and relative-value strategies; quantitative portfolio construction, scenario analysis, and institutional risk management.

Natixis — Proprietary / Quantitative Trading (London / Paris): Systematic credit and relative-value research across derivatives and capital structures.

J.P. Morgan Chase (Asia): Rates, swaps, options, foreign-exchange and cross-currency relative-value strategies.

Morgan Stanley (Tokyo): Institutional fixed-income derivatives and structured interest-rate products.

Salomon Brothers (Tokyo): Nikkei index cash-and-carry arbitrage based on futures–spot dislocations.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence — M.S., Natural Language Processing (2025); thesis: On Why Form Shapes Reason.

Stanford University — M.S., Financial Mathematics | **Massachusetts Institute of Technology** — B.S., Finance

École 42, Paris — Computer science program | **Waseda University** — Graduate study, Investment Management

TECHNICAL SKILLS

Python, PyTorch, JAX, Google ADK, Gemini, Git, pytest, Google Cloud (Cloud Run, Pub/Sub, Firestore); agent orchestration, retrieval, RL, probabilistic modeling, experiment design, ablations, evaluators, verification, uncertainty analysis.